DATABASE MANAGEMENT LAB

LAB Report

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

K.R Mangalam University

Submitted by:

Nakul Arora

(2401010025)

Submitted to

Ms. Mansi Kajal

Department of Computer Science and Engineering

School of Engineering and Technology

K.R Mangalam University, Gurugram- 122001, India

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Experiment : 1

Question 1. Consider following databases and draw ER diagram and convert entities

and relationships to relation table for a given scenario:

COLLEGE DATABASE

STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

Solution:

1. The entities **STUDENT, SEMSEC, SUBJECT, and IAMARKS** are identified and represented using rectangles.
2. The attributes are represented using ellipses:

o **STUDENT:** USN (Primary Key), SName, Address, Phone, Gender
♣ USN is underlined as it is the Primary Key.
♣ Phone is shown as a single-valued attribute.

o **SEMSEC:** SSID (Primary Key), Sem, Sec
♣ SSID is underlined as the Primary Key.

o **SUBJECT:** Subcode (Primary Key), Title, Sem, Credits
♣ Subcode is underlined as the Primary Key.

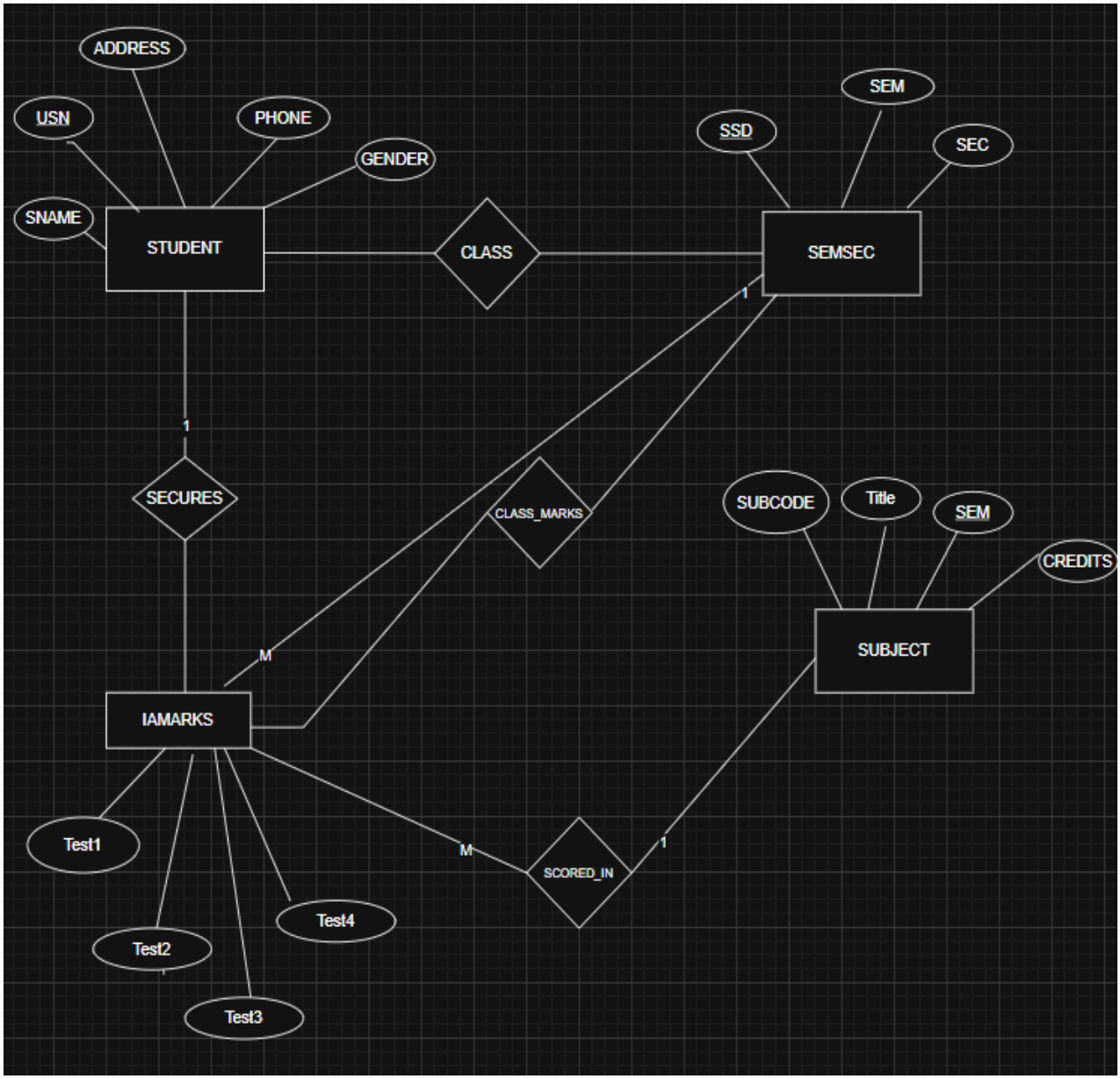
o **IAMARKS:** USN, Subcode, SSID, Test1, Test2, Test3, FinalIA
♣ USN, Subcode, and SSID together form a composite key.
♣ Test1, Test2, Test3, and FinalIA are normal attributes.

3. Primary keys are represented by underlining the attribute names.
4. **CLASS** is drawn as a relationship (diamond shape) between **STUDENT** and **SEMSEC**.
5. The relationship between **STUDENT and SEMSEC** through CLASS is **M : 1**, meaning many students belong to one semester and section.
6. The **M : N relationship between STUDENT and SUBJECT** is resolved using the associative entity **IAMARKS**.
7. The relationship **STUDENT – IAMARKS** is **1 : M**, indicating one student can have multiple internal assessment records.
8. The relationship **SUBJECT – IAMARKS** is **1 : M**, indicating one subject can have multiple IA records for different students.

9. The relationship **SEMSEC – IAMARKS** is **1 : M**, showing that marks are maintained semester-section wise.

10. All entities, attributes, relationships, primary keys, foreign keys, and cardinalities are clearly represented in the ER diagram.

ER DIAGRAM:



Experiment : 2

Question 2. Consider following databases and draw ER diagram and convert entities

and relationships to relation table for a given scenario:

COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Salary, Sex, SuperSSN, DNO)
DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate)
PROJECT (PNo, PName, PLocation, DNO)
DLOCATION (DNo, DLoc)
WORKS_ON (SSN, PNo, Hours)
MANAGES (SSN, DNo)

Solution:

1. The entities **EMPLOYEE, DEPARTMENT, PROJECT, and DLOCATION** are identified and represented using rectangles.
2. The attributes are represented using ellipses:

o **EMPLOYEE:** SSN (Primary Key), Name, Address, Salary, Sex, SuperSSN, DNO
♣ SSN is underlined as it is the Primary Key.
♣ SuperSSN is a foreign key referencing EMPLOYEE (self-relationship).
♣ DNO is a foreign key referencing DEPARTMENT.

o **DEPARTMENT:** DNo (Primary Key), DName, MgrSSN, MgrStartDate
♣ DNo is underlined as the Primary Key.
♣ MgrSSN is a foreign key referencing EMPLOYEE.

o **PROJECT:** PNo (Primary Key), PName, PLocation, DNO
♣ PNo is underlined as the Primary Key.
♣ DNO is a foreign key referencing DEPARTMENT.

o **DLOCATION:** DNo, DLoc
♣ DNo and DLoc together form a composite key.
♣ DNo is a foreign key referencing DEPARTMENT.

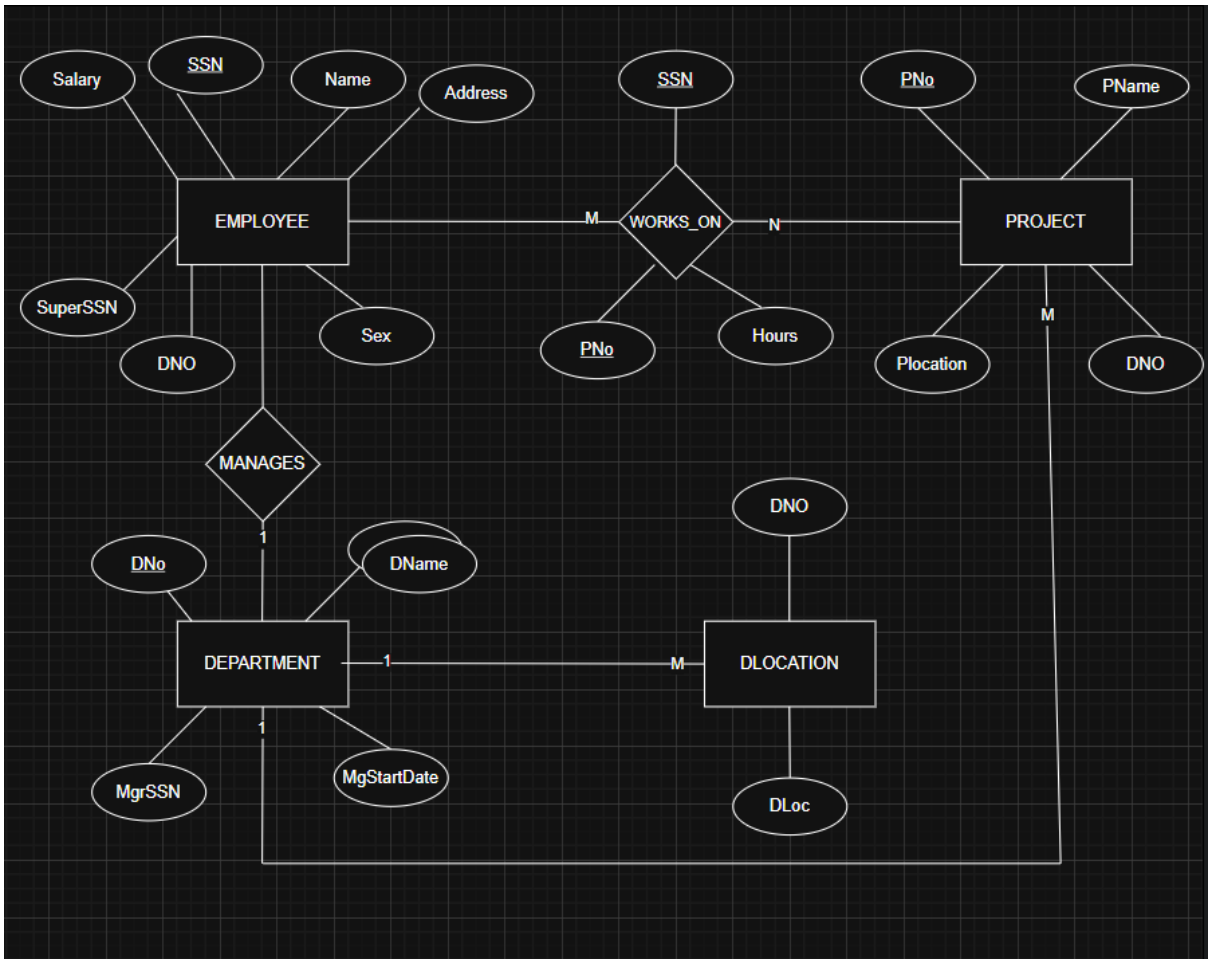
o **WORKS_ON:** SSN, PNo, Hours
♣ SSN and PNo together form a composite key.
♣ Hours is a descriptive attribute of the relationship.

3. Primary keys are represented by underlining the attribute names.
4. The relationship **WORKS_ON** is drawn as a diamond between **EMPLOYEE** and **PROJECT** with **M : N cardinality**, indicating that:
    - One employee can work on many projects.
    - One project can have many employees.

- Hours attribute represents the number of hours worked.

5. The relationship **MANAGES** is drawn between **EMPLOYEE** and **DEPARTMENT** with **1 : 1 cardinality**, indicating:
    - One employee manages one department.
    - Each department has one manager.
6. The relationship between **EMPLOYEE and DEPARTMENT** (via DNO) is **M : 1**, meaning:
    - Many employees work in one department.
7. The relationship between **DEPARTMENT and PROJECT** is **1 : M**, meaning:
    - One department controls many projects.
    - Each project belongs to one department.
8. The relationship between **DEPARTMENT and DLOCATION** is **1 : M**, meaning:
    - One department can have multiple locations.
9. A recursive relationship exists in EMPLOYEE through **SuperSSN**, representing that:
    - One employee can supervise many employees.
10. All entities, attributes, relationships, primary keys, foreign keys, and cardinalities are clearly represented in the ER diagram.

ER DIAGRAM:



Experiment : 3

Question : Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name: string, price: integer)

Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

For the above schema, perform the following—

1. Create the tables with the appropriate integrity constraints.

Output:

```
CREATE DATABASE CUSTOMER_SALE;
USE CUSTOMER_SALE;
CREATE TABLE CUSTOMER(
CUST_ID INT PRIMARY KEY,
CUST_NAME VARCHAR(100)
);
CREATE TABLE ITEM (
ITEM_ID INT PRIMARY KEY,
ITEM_NAME VARCHAR(100),
PRICE INT);
CREATE TABLE SALE (
BILL_NO INT ,
BILL_DATA DATE,
CUST_ID INT,
ITEM_ID INT,
QTY_SOLD INT ,
FOREIGN KEY (CUST_ID) REFERENCES CUSTOMER(CUST_ID),
FOREIGN KEY (ITEM_ID) REFERENCES ITEM(ITEM_ID)
);
```

2. Insert around 10 records in each of the tables.

```
INSERT INTO
CUSTOMER(CUST_ID ,CUST_NAME)
VALUES
(1,'nakul'),
(2,'Harsh'),
(3,'Ankit'),
(4,'Aryan'),
(5,'Hridyanshu'),
(6,'Ashok'),
(7,'Alok'),
(8,'Aman'),
(9,'Ashish'),
(10,'Mohit');
```

CUST_ID	CUST_NAME
1	Nakul
2	Harsh
3	Ankit
4	Aryan
5	Hridyanshu
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
11	RAHUL
NULL	NULL

3.

```sql
INSERT INTO ITEM VALUES
(101,'Pen',50),
(102,'Notebook',120),
(103,'Bag',800),
(104,'Bottle',250),
(105,'Shoes',1500),
(106,'Watch',2200),
(107,'Mouse',500),
(108,'Keyboard',700),
(109,'Charger',300),
(110,'Headphones',1800);
SELECT * FROM ITEM;
```

ITEM_ID	ITEM_NAME	PRICE
101	Pen	50
102	Notebook	120
103	Bag	800
104	Bottle	250
105	Shoes	1500
106	Watch	2200
107	Mouse	500
108	Keyboard	700
109	Charger	300
110	Headphones	1800
NULL	NULL	NULL

```sql
INSERT INTO SALE VALUES
(1,CURDATE(),1,103,1),
(2,CURDATE(),2,101,3),
(3,CURDATE(),2,105,2),
(4,'2026-02-15',3,104,1),
(5,'2026-02-14',4,106,1),
(6,'2026-02-13',5,102,4),
(7,CURDATE(),6,110,1),
(8,'2026-02-16',7,108,2),
(9,CURDATE(),5,109,5),
(10,'2026-02-12',8,107,1);
SELECT * FROM SALE;
```

BILL_NO	BILL_DATA	CUST_ID	ITEM_ID	QTY_SOLD
1	2026-02-23	1	103	1
2	2026-02-23	2	101	3
3	2026-02-23	2	105	2
4	2026-02-15	3	104	1
5	2026-02-14	4	106	1
6	2026-02-13	5	102	4
7	2026-02-23	6	110	1
8	2026-02-16	7	108	2
9	2026-02-23	5	109	5
10	2026-02-12	8	107	1

4. List all the bills for the current date with the customer names and item numbers

```sql
SELECT S.BILL_NO, C.CUST_NAME, S.ITEM_ID
FROM SALE S
JOIN CUSTOMER C ON S.CUST_ID = C.CUST_ID
WHERE S.BILL_DATA = CURDATE();
```

BILL_NO	CUST_NAME	ITEM_ID
1	Nakul	103
2	Harsh	101
3	Harsh	105
3	Harsh	105
7	Ashok	110
9	Aman	110

5. List the total Bill details with the quantity sold, price of the item and the final amount.

```
SELECT
    S.BILL_NO,
    C.CUST_NAME,
    I.ITEM_NAME,
    S.QTY_SOLD,
    I.PRICE,
    (S.QTY_SOLD * I.PRICE) AS FINAL_AMOUNT
FROM SALE S
JOIN CUSTOMER C ON S.CUST_ID = C.CUST_ID
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID;
```

BILL_NO	CUST_NAME	ITEM_NAME	QTY_SOLD	PRICE	FINAL_AMC
1	Nakul	Bag	1	800	800
2	Harsh	Pen	3	101	150
3	Harsh	Shoes	2	1500	3000
4	Ankit	Bottle	1	250	250
5	Aryan	Watch	1	2200	2200
6	Hridyanshu	Notebook	4	120	480
7	Ashok	Headphones	1	1800	1800
8	Alok	Keyboard	2	700	1400
9	Hridyanshu	Charger	5	300	1500
10	Aman	Mouse	1	500	500

6. List the details of the customer who have bought a product which has a price>200

```
SELECT DISTINCT C.*
FROM CUSTOMER C
JOIN SALE S ON C.CUST_ID = S.CUST_ID
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE I.PRICE > 200;
```

CUST_ID	CUST_NAME
1	Nakul
3	Ankit
2	Harsh
4	Aryan
8	Aman
7	Alok
5	Hridyanshu
6	Ashok

7. Give a count of how many products have been bought by each customer

```
SELECT C.CUST_NAME, SUM(S.QTY_SOLD) AS TOTAL_PRODUCTS_BOUGHT
FROM CUSTOMER C
JOIN SALE S ON C.CUST_ID = S.CUST_ID
GROUP BY C.CUST_NAME;
```

CUST_NAME	TOTAL_PRODUCTS_BOUGHT
Nakul	1
Harsh	5
Ankit	1
Aryan	1
Hridyanshu	9
Ashok	1
Alok	2
Aman	1

8. Give a list of products bought by a customer having cust_id as 5.

```
SELECT I.ITEM_NAME, S.QTY_SOLD
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE S.CUST_ID = 5;
```

ITEM_NAME	QTY_SOLD
Notebook	4
Charger	5

z

9. List the item details which are sold as of today.

```
SELECT I.*
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE S.BILL_DATA = CURDATE();
```

ITEM_ID	ITEM_NAME	PRICE
103	Bag	800
101	Pen	50
105	Shoes	1500
110	Headphones	1800
109	Charger	300

10. Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold,Amount

```
CREATE VIEW Bill_Details AS
SELECT
    S.BILL_NO,
    S.BILL_DATA,
    S.CUST_ID,
    S.ITEM_ID,
    I.PRICE,
    S.QTY_SOLD,
    (S.QTY_SOLD * I.PRICE) AS AMOUNT
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID;

SELECT * FROM BILL_DETAILS;
```

BILL_NO	BILL_DATA	CUST_ID	ITEM_ID	PRICE	QTY_SOLD	AMOUNT
1	2026-02-23	1	103	800	1	800
2	2026-02-23	2	101	50	3	150
3	2026-02-23	2	105	1500	2	3000
4	2026-02-15	3	104	250	1	250
5	2026-02-14	4	106	2200	1	2200
6	2026-02-13	5	102	120	4	480
7	2026-02-23	6	110	1800	1	1800
8	2026-02-16	7	108	700	2	1400
9	2026-02-23	5	109	300	5	1500
10	2026-02-12	8	107	500	1	500

11. Create a view which lists the daily sales date wise for the last one week

```
CREATE VIEW WEEKLY_SALES AS
SELECT
    BILL_DATA,
    SUM(S.QTY_SOLD * I.PRICE) AS TOTAL_DAILY_SALES
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE BILL_DATA >= CURDATE() - INTERVAL 7 DAY
GROUP BY BILL_DATA;

SELECT * FROM WEEKLY_SALES;
```

BILL_DATA	TOTAL_DAILY_SALES
2026-02-23	7250
2026-02-16	1400

Experiment: 4

Question: Consider the following table: Table: CLASS (ID,Name)

```
CREATE DATABASE NAKUL;
USE NAKUL;
CREATE TABLE CLASS(
ID INT,
NAME VARCHAR(100));
INSERT INTO
CLASS(ID, NAME)
VALUES
(1,'Nakul'),
(2,'Harsh'),
(3,'Ankit'),
(4,'Aryan'),
(5,'Hridyanshu'),
(6,'Ashok'),
(7,'Alok'),
(8,'Aman'),
(9,'Ashish'),
(10,'Mohit');
SELECT * FROM CLASS;
```

ID	NAME
1	Nakul
2	Harsh
3	Ankit
4	Aryan
5	Hridyanshu
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit

1. SET AUTOCOMMIT = 0; and START TRANSACTION;

```
SET autocommit = 0;
START TRANSACTION;
```

2. INSERT INTO class VALUES (11,'Rahul');

```
INSERT INTO CLASS VALUES (11,'RAHUL');
SELECT * FROM CLASS;
```

ID	NAME
1	Nakul
2	Harsh
3	Ankit
4	Aryan
5	Hridyanshu
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
11	RAHUL
12	CHRIS
13	BRAVO
11	RAHUL

3. UPDATE class SET name = ‘Abhijeet’ WHERE id= ‘5’;

```
26 •  SET SQL_SAFE_UPDATES = 0;
27 •  UPDATE CLASS SET NAME = 'ABHIJEET' WHERE ID = 5;
28 •  SELECT * FROM CLASS;
```

	ID	NAME
▶	1	Nakul
	2	Harsh
	3	Ankit
	4	Aryan
	5	ABHIJEET
	6	Ashok
	7	Alok
	8	Aman
	9	Ashish
	10	Mohit
	11	RAHUL

4. SAVEPOINT A:

```
28 •  SAVEPOINT A;
```

5. INSERT INTO class VALUES (12, ‘Chris’);

```
30 •  INSERT INTO CLASS VALUES (12,'CHRIS');
```

	ID	NAME
▶	1	Nakul
	2	Harsh
	3	Ankit
	4	Aryan
	5	ABHIJEET
	6	Ashok
	7	Alok
	8	Aman
	9	Ashish
	10	Mohit
	11	RAHUL
	12	CHRIS

6. SAVEPOINT B;

```
31 •  SAVEPOINT B;
```

7. INSERT INTO class VALUES (13, ‘Bravo’);

```
32 •  INSERT INTO CLASS VALUES (13,'BRAVO');
33 •  SELECT * FROM CLASS;
```

	ID	NAME
▶	1	Nakul
	2	Harsh
	3	Ankit
	4	Aryan
	5	ABHIJEET
	6	Ashok
	7	Alok
	8	Aman
	9	Ashish
	10	Mohit
	11	RAHUL
	12	CHRIS
	13	BRAVO

8. SAVEPOINT C;

```
34 •    SAVEPOINT C;
```

9. ROLLBACK TO B;

```
36 •    ROLLBACK TO B;
```

	ID	NAME
▶	1	Nakul
	2	Harsh
	3	Ankit
	4	Aryan
	5	ABHIJEET
	6	Ashok
	7	Alok
	8	Aman
	9	Ashish
	10	Mohit
	11	RAHUL
	12	CHRIS

10. ROLLBACK TO A;

```
38 •    ROLLBACK TO A;
39 •    SELECT * FROM CLASS;
```

	ID	NAME
▶	1	Nakul
	2	Harsh
	3	Ankit
	4	Aryan
	5	ABHIJEET
	6	Ashok
	7	Alok
	8	Aman
	9	Ashish
	10	Mohit
	11	RAHUL

11. COMMIT;

```
40 •    COMMIT;
```

Experiment: 4.2

Consider the following table: Table: ACCOUNT_TXN

Acc_ID Holder_Name Balance

101	**Aman**	**5000**
102	**Bhavna**	**7000**
103	**Chirag**	**6000**

```sql
CREATE TABLE ACCOUNT_TXN(
ACC_ID INT,
HOLDER_NAME VARCHAR(100),
BALANCE INT
);
INSERT INTO ACCOUNT_TXN
VALUES
(101,'AMAN',5000),
(102,'BHAVNA',7000),
(103,'CHIRAG',6000);
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	5000
102	BHAVNA	7000
103	CHIRAG	6000

Perform the following TCL-based SQL script and write the output after each SELECT statement:

1. INSERT INTO ACCOUNT_TXN VALUES (104, 'Diya', 4000);

```sql
INSERT INTO ACCOUNT_TXN
VALUES
(104,'DIYA',4000);
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	5000
102	BHAVNA	7000
103	CHIRAG	6000
104	DIYA	4000

2. SAVEPOINT S1

```sql
SET autocommit = 0;
START TRANSACTION;
SAVEPOINT S1;
```

3. UPDATE ACCOUNT_TXN SET Balance = Balance + 1500 WHERE Acc_ID = 101

```sql
UPDATE ACCOUNT_TXN SET BALANCE = BALANCE+1500 WHERE ACC_ID = 101;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	6500
102	BHAVNA	7000
103	CHIRAG	6000
104	DIYA	4000

4. **UPDATE ACCOUNT_TXN SET Balance = Balance - 1000 WHERE Acc_ID = 102**

```
UPDATE ACCOUNT_TXN SET BALANCE = BALANCE - 1000 WHERE ACC_ID = 102;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	8000
102	BHAVNA	6000
103	CHIRAG	6000
104	DIYA	4000

5. **SAVEPOINT S2;**

```
SAVEPOINT S2;
```

6. **DELETE FROM ACCOUNT_TXN WHERE Acc_ID = 103**

```
DELETE FROM ACCOUNT_TXN WHERE ACC_ID = 103;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	8000
102	BHAVNA	6000
104	DIYA	4000

7. **INSERT INTO ACCOUNT_TXN VALUES (105, 'Eshan', 3000);**

```
INSERT INTO ACCOUNT_TXN
VALUES
(105,'ESHAN',3000);
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	8000
102	BHAVNA	6000
104	DIYA	4000
105	ESHAN	3000

8. **SAVEPOINT S3**

```
SAVEPOINT S3;
```

9. **UPDATE ACCOUNT_TXN SET Holder_Name = 'Bhavna Sharma' WHERE Acc_ID = 102;**

```
UPDATE ACCOUNT_TXN SET HOLDER_NAME = 'BHAVNA SHARMA' WHERE ACC_ID = 102;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	5000
102	BHAVNA SHARMA	7000
103	CHIRAG	6000
104	DIYA	4000

11. ROLLBACK TO S2;

```
ROLLBACK TO S2;
```

	ACC_ID	HOLDER_NAME	BALANCE
▶	101	AMAN	6500
	102	BHAVNA SHARMA	6000
	103	CHIRAG	6000
	104	DIYA	4000

12. INSERT INTO ACCOUNT_TXN VALUES (106, 'Farah', 4500)

```
INSERT INTO ACCOUNT_TXN
VALUES
(106,'FARAH',4500);
SELECT * FROM ACCOUNT_TXN;
```

	ACC_ID	HOLDER_NAME	BALANCE
▶	101	AMAN	6500
	102	BHAVNA SHARMA	6000
	103	CHIRAG	6000
	104	DIYA	4000
	104	DIYA	4000
	106	FARAH	4500

13. SAVEPOINT S4;

```
SAVEPOINT S4;
```

14. UPDATE ACCOUNT_TXN SET Balance = Balance + 500 WHERE Acc_ID IN (101,104);

SELECT * FROM ACCOUNT_TXN

```
UPDATE ACCOUNT_TXN SET BALANCE = BALANCE + 500 WHERE ACC_ID IN (101,104);
SELECT * FROM ACCOUNT_TXN;
```

	ACC_ID	HOLDER_NAME	BALANCE
▶	101	AMAN	7000
	102	BHAVNA SHARMA	6000
	103	CHIRAG	6000
	104	DIYA	4500
	104	DIYA	4500
	106	FARAH	4500

15. ROLLBACK TO S4;

SELECT * FROM ACCOUNT_TXN;

```
ROLLBACK TO S4;
SELECT * FROM ACCOUNT_TXN;
```

	ACC_ID	HOLDER_NAME	BALANCE
▶	101	AMAN	6500
	102	BHAVNA SHARMA	6000
	103	CHIRAG	6000
	104	DIYA	4000
	104	DIYA	4000
	106	FARAH	4500

4000

16. COMMIT

```
COMMIT;
```

17. Identify which rows are permanently saved after COMMIT.

	ACC_ID	HOLDER_NAME	BALANCE
▶	101	AMAN	6500
	102	BHAVNA SHARMA	6000
	103	CHIRAG	6000
	104	DIYA	4000
	104	DIYA	4000
	106	FARAH	4500